COOLSHADE · RISK REGISTER AND HAZARD ANALYSIS

CoolShade risk register and hazard analysis

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DOCUMENT CONTROL

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REVISION HISTORY

Version	Date	Author	Change
0.1	2026-10-30	Amish Chadha	First issue in the kit 1.9.1 upgrade: project risks, preliminary hazard analysis and a first-pass design FMEA with RPN for option A, from CSH-CAL-001 v0.6 and CSH-STM-001

CoolShade risk register and hazard analysis

The top project risk is that no site or partner agrees to a public trial, because a misting system needs a hygiene plan that a health authority accepts (P1, P3). The top hazard is *Legionella* from the mist (H1), controlled by water that is never older than 24 h, a line that drains within 30 s of a session, a filter and weekly disinfection, with a residual that stays high until a partner lab samples the water at TRL 4. The highest design FMEA scores are a blocked vacuum breaker or wet runs that leave warm water in the line (RPN 48), a shade cloth left on in an unforecast gust (RPN 48), anchors that pull out of an unknown slab (RPN 40), and a drifting or misted humidity sensor (RPN 36 each). Method: STANDARDS section 23 and ISO 12100. Likelihood, impact, severity and occurrence are scored 1 (low) to 5 (high); detection 1 (certain to be caught) to 5 (unlikely to be caught).

Project risk register

Table 1. Project risks.

#	Risk	Cause	Effect	Likelihood	Impact	Owner (role)	Response
P1	No site or partner agrees to a trial	Candidate named, nothing agreed (CSH-DEC-001); misting on public land raises hygiene and liability questions	Requirements and the reference site stay desk assumptions; no TRL 4	3	5	Amish	Approach the market operator and a university heat group with the hygiene plan first
P2	Parts cost stays over the target	USD 882 against USD 775; steel prices look low until quoted	Prototype not affordable for a community group	4	3	Design lead	Savings in CSH-DEC-001 (base plates, steel panel rails); steel quotes; budget raise only proposed, never made here
P3	Health authority will not accept public misting	*Legionella* risk in 34 °C tank water (STEM S14)	No public trial; option B or C becomes the fallback	3	5	Market operator	Hygiene plan with named responsible person; sampling with a partner lab; option B has no free aerosol (CSH-OPT-001)
P4	Nozzle maker has no droplet data or the droplets are too big	Dv0.9 not published for the 0.4 mm swirl nozzle at 7 bar	Wetting (PR-005); complaints	3	4	Design lead	Ask makers for Dv0.9 and flow before buying (CSH-DEC-001 to-confirm item 1); option B as fallback
P5	People do not want to stand in a mist	Wet clothes, hair and hygiene concerns	Low use	3	3	Partner	Ask in co-design (CSH-PRB-001 open questions) before building
P6	Theft or vandalism of the panel, battery, pump or fabric	Public site, no staff	Loss of function; cost	3	3	Market operator	Lockable enclosure and tamper-proof screws; bolted frame; fabric removable and stored off site between storms
P7	Design gust at the site above 30 m/s	Storm climate not the hot, dry design case	Frame and anchors under-designed	2	5	Design lead	Site wind data per site (CSH-REG-001 G3); fabric off above 15 m/s

Preliminary hazard analysis

Table 2. Hazards. Residual is the severity x likelihood score (1 to 25) after the control.

#	Hazard	Cause	Harm	Severity	Likelihood	Control	Residual
H1	*Legionella* from inhaled mist	Warm tank water (about 34 °C) and a line that stands wet	Legionnaires' disease or Pontiac fever	5	2	5 um filter; water never older than 24 h; line drains within 30 s (vacuum breaker); opaque shaded lockable tank; weekly disinfection; a named responsible person	10
H2	Frame or roof lifted or blown over in a gust	Wind uplift, anchors short, cloth left fitted	Crush injury; damage	5	2	Cloth off above 15 m/s forecast; M16 anchors; frame checks 0.14 utilization at 30 m/s with the cloth off (PR-010)	10
H3	Shade cloth torn loose in a storm	Cloth fitted in a gust above 15 m/s	Flying cloth; eyelet and cord strike	3	3	Fit only when forecast gusts are below 15 m/s; reinforced hem and eyelets; removable in 15 min	6
H4	Electric shock or fire	12 V DC battery and wiring; short circuit	Burn; fire	4	1	12 V only; 15 A terminal fuse; LiFePO4 with BMS; IP65 lockable enclosure	4
H5	Slip on wet ground	Mist or drips wet the pavement	Fall and injury	3	3	Droplets sized to evaporate (PR-005, not yet verified); misting only in hot, dry air; no puddles as a requirement	6
H6	Head strike	Knee brace ends and nozzles at 2.22 m	Cut or bump	2	2	Clear height 2,223 mm or more (PR-002); flattened brace ends; edges broken	3
H7	Cuts and pinches in assembly	Sharp section ends, rivet nut tool, cloth lacing	Cuts, pinched fingers	2	3	End caps on tubes; edges broken in the model; gloves in the build plan	4
H8	Tank tips or empties onto people	128 kg full; an empty tank tips in 30 m/s	Crush; spill	4	2	Ratchet strap and two stop cleats anchored to the slab (CSH-CAL-001 G12)	6
H9	Manual handling	Posts of 21.0 kg; panel 7 kg	Back injury	3	3	Two-person lifts; no part over 40 kg (PR-017)	4
H10	Mist reduces visibility or wets the road	Street-side posts at the kerb	Traffic incident	3	2	Nozzles inboard of the long beams; setback check at the site (CSH-DEC-001 to-confirm item 10)	4
H11	Hot surfaces and sun on metal	Pump at 60 W; steel in full sun	Burn on contact	2	2	Pump in a vented box on the plate; cloth shades the frame	2

#	Hazard	Cause	Harm	Severity	Likelihood	Control	Residual
H12	Falling solar panel	7 kg panel on rails and L-feet on a 7 degree roof	Strike injury	4	1	Four L-feet with M8 bolts into rivet nuts, four end clamps (pull per rivet nut about 0.16 kN, CSH-CAL-001 G15)	4

Design FMEA (first pass, P3)

Table 3. Failure modes. RPN = severity x occurrence x detection.

#	Item	Function	Failure mode	Effect	Cause	S	O	D	RPN	Action
F1	Vacuum breaker (air admittance valve)	Let air in so the mist line drains	Valve blocked or stuck shut	Line stays full of warm water between sessions	Dust, insects, scale; the horizontal runs and the nozzle bodies stay wet even when it works	4	3	4	48	Cap with a mesh; inspect weekly; check the drain time and a dry line at every visit; fall the long runs toward the riser (to assess); high priority for the hygiene plan
F2	M16 anchors	Hold the posts to the slab	Pull out or crack the slab	Frame uplift in a gust	Unknown slab, short embedment	5	2	4	40	Pull test each anchor at installation; site engineer check (CSH-REG-001 G6, STEM S20)
F3	Humidity and temperature sensor in the shield	Decide when to mist	Reading drifts or wets	Mists in humid air or not at all	Condensate, dust, ageing	3	3	4	36	Plate shield; replace on a schedule; sanity check against a second reading in firmware (TRL 4)
F4	Rivet nuts in 2 mm tube	Fix the roof joints	Pull out	A loose roof joint	Thin wall, vibration, over-torque	4	2	4	32	Maker's pull-out rating check (STEM S21); torque limit in the build plan
F5	Normally open drain valve	Open when the pump stops	Stuck closed	Line stays full	Debris, coil failure	4	2	4	32	Fail-safe normally open type; clean strainer; test at every visit
F6	Shade cloth and eyelets	Cover the roof	Eyelet tears, hem frays	Cloth flaps loose	UV, wind	3	3	3	27	Reinforced hem; replace yearly; fit only below 15 m/s
F7	PIR presence sensor	Detect people	Misses a person standing still in 38 to 45 °C air	No mist when someone waits	Low contrast near body temperature (STEM S15)	2	4	3	24	Sensor datasheet derating; test at TRL 4; radar alternative to Amish (CSH-DEC-001 D10)

#	Item	Function	Failure mode	Effect	Cause	S	O	D	RPN	Action
F8	Enclosure gasket and glands	Keep the electronics dry	Water or dust gets in	Short circuit or corrosion	Gasket set, gland not tightened	4	2	3	24	IP65 enclosure; moulded mounting holes outside the gasket (CSH-DDR-003); check glands at build
F9	Aluminium rails on galvanised feet	Carry the panel	Galvanic corrosion at the contact	Loosening	Zinc and aluminium in a wet, salty air	2	3	4	24	Isolating washers to assess; inspect yearly
F10	Tank strap	Stop the tank tipping	Strap fails or slackens	Tank tips in a gust	UV, ratchet creep	3	2	3	18	Two stop cleats as a second restraint; check the strap at each refill
F11	Nozzles, 0.4 mm	Make the mist	Scale blocks the orifice	Less cooling, uneven spray	Hard water (STEM S12)	2	4	2	16	Tool-free removal; descale; 5 um filter; water hardness check
F12	Pump	Pressurise the line	Runs dry, or the diaphragm wears from about 450 starts a day (20 s on, 20 s off)	No mist	Low tank, cycling, inrush	2	3	2	12	Low-level float switch cuts the pump; spare diaphragm; fewer, longer cycles to assess at TRL 4
F13	Battery charge cutoff	Protect the cells	Charging pauses in air above about 42.7 °C in the cloth's shade, about 38 °C with the enclosure in direct late sun	Less charge on the hottest days; cells age faster at 40 to 47 °C	Cell charge limit of about 45 °C (STEM S11)	2	4	3	24	White enclosure (CSH-DEC-001 D9); enclosure sun exposure and a sunshield to assess; 1.26 design days of reserve at beginning of life
F14	Shade cloth, fitted	Shade	Cloth left on in an unforecast gust	Beams loaded to their limit near 16 m/s (19.6 m/s unfactored); at	Dust storms and microbursts in hot, dry climates come	4	3	4	48	Fit only when forecast gusts are below 15 m/s (CSH-DDR-002); cord lacing that releases under

#	Item	Function	Failure mode	Effect	Cause	S	O	D	RPN	Action
				30 m/s utilization 3.5	with little warning					load is studied at TRL 4 (CSH-DDR-002 N1c); site wind data
F15	Sensor shield and PIR window	Read the air and the people	Mist reaches the humidity sensor or the PIR window	False humid reading, mist off in dry air, or false presence	Drift of droplets in a breeze	3	3	4	36	Plate shield; sensor head 375 mm outboard of the front right post (CSH-DDR-003); check against nozzle drift at TRL 4
F16	Drain valve coil	Open the line when the pump stops	Valve opens in mid-session if its 5 W hold is lost	Session stops early; water drains to the slab	Power dip, loose connector	2	3	2	12	Fail-safe normally open type is kept (CSH-DDR-002); locking connector

Sources

- World Health Organization. "Legionellosis." https://www.who.int/news-room/fact-sheets/detail/legionellosis (checked 2026-10-30).
- CSH-CAL-001 and CSH-STM-001 for the loads, times and energy quoted.
- ISO 12100, general principles for risk assessment (method only; text not consulted).